

Nakul Chamariya

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TECHNICAL SKILLS AND TOOLS

Languages: Python, TensorFlow, PyTorch

Data Science: Preprocessing, Feature Engineering, Visualization, Statistical Analysis

Machine Learning: Supervised, Unsupervised, Scikit-learn, XGBoost, LightGBM

Deep Learning: TensorFlow, PyTorch, Keras, OpenCV, Transformers

Gen-AI & NLP: GPT-4, LLAMA 3, Langchain, Crew-ai, Hugging Face, Stable Diffusion, OpenAI Whisper

Computer Vision: Object Detection, Segmentation, OCR, OpenCV, Pillow, Yolo

MLOps & DevOps: Docker, DVC, CI/CD (GitHub Actions, CircleCI), AWS (EC2, S3, Lambda, ECR, ECS)

Web: Flask, FastAPI, Streamlit, Selenium, Beautiful Soup

Databases: MySQL, MongoDB, SQLite, Pinecone

Other: Git, Linux

WORK EXPERIENCE

MySellerCentral

Feb'24 – Till date

Senior Data Scientist

Mumbai, India

Skills Used: AWS, Deep Learning, Computer Vision, NLP, Gen-AI, Web Scraping, MLOps, API

- **AI for Marketplace Optimization**

- Developed and deployed **AI-based content generation modules** on **AWS ECS Fargate**, enabling sellers to upload audio or text in local languages to create **optimized content** (titles, descriptions, keywords) for multiple marketplaces such as **Amazon, Flipkart, and Shopify**. Implemented **autoscaling, load balancing, and CI/CD workflows** for seamless deployment.
- Built **content grading modules** to assess and provide improvement suggestions for marketplace listings. Created a **multimodal multi-agent Gen-AI system** for automated listing checks (image and text) catering to **over 2 million sellers**, ensuring seamless integration with various marketplaces like Amazon and ONDC.
- Used a **Gen-AI scraper** libraries to extract product pages and reviews from Amazon, enhancing the 360-degree customer experience. Implemented **attribute-based sentiment analysis** to evaluate product attributes from customer reviews, reducing prediction time from **20 minutes to 2 minutes** through **multithreading and multiprocessing** with LLM API calls.

- **AI for Media Creation**

- Engineered AI-based modules for generating **infographics, lifestyle images, and videos** for sellers, deployed on **serverless GPU instances**, resulting in a monthly **cost savings of over \$1,000**.

- **AI for Seller Insights and Support**

- Created an **SQL-based sales analytics module** that responds to audio queries in various local languages, providing data-driven insights with visualizations and detailed reports.
- Developed a **RAG-based chatbot** to answer queries for sellers, fine-tuned **LLaMA 3 8B** on company-specific data, and deployed it on **AWS SageMaker** to efficiently handle seller queries.

- **Alerts and Notification Systems**

- Developed **AWS SNS and Lambda functions** to automate notifications for API key exhaustion (OpenAI, Keepa, Zyte) and enhanced security by using **AWS Secrets Manager** to eliminate exposure of sensitive access credentials.

- **Efficient Resource Management**

- Implemented queue-based request handling using **Celery and SQS** to prevent server-side bottlenecks, and integrated **Ray** for enhanced resource utilization in both production and development environments.

- **Infrastructure and CI/CD Development**

- Led the setup and management of production and development environments, including **GitHub repositories** with stage, dev, and prod branches. Developed CI/CD pipelines using **Jenkins and GitHub Actions** for seamless deployment across environments.
- Implemented containerization for all modules using **Docker**, optimizing with **multi-stage builds** and **Alpine images**, achieving up to **5x cost savings and 10x latency improvements**.

- **AI Model Deployment and Optimization**

- Deployed and managed all AI modules on **AWS Lambda** for cost optimization and load handling, and created CI/CD pipelines for the same. Utilized **model quantization** wherever possible to reduce costs and inference latency.
- Extensively leveraged **Bedrock and OpenAI GPT** for module development, implementing end-to-end pipelines with **comprehensive monitoring and alert systems** with **Grafana and Prometheus** at both instance and module levels to ensure high availability and performance.

goGlocal Private Limited

Data Scientist

Feb'23 – Feb'24

Mumbai, India

Skills Used: AWS, Deep Learning, Computer Vision, NLP, Gen-AI, Web Scraping, MLOps, API

- Built **web scraping pipelines** and a **real-time price tracker** with **automated email alerts** for actionable insights.
- Designed tools integrating **OCR, object detection**, and **Gen-AI** for optimizing product listings and images.
- Developed **sentiment analysis**, **sales forecasting**, and **bidding optimization** models for enhanced marketplace performance.
- Engineered a **multilingual translation tool** for SKUs and **image translation** with OCR for compliance and localization.
- Enhanced chatbots using **Llama2, Langchain**, and **conversation history** modules for compliance and e-commerce support.
- Created **APIs** with **Celery and Redis**, along with setting up a **CI/CD pipeline** for scalable deployment.
- Designed a **strategic insights dashboard** to provide category-level marketplace analytics.

Edvak Technologies INC

Data Scientist

Dec'21 – Jan'23

Hyderabad, India

Skills Used: AWS, Machine Learning, Deep Learning, NLP, Speech-to-Text, MLOps, API

- Revolutionized **lab test and medication recommendations** with a **multi-label classification** system.
- Enhanced text accuracy with **Diagnosis Prediction** using **Clinical-Bert** and **Clinical-NER** for medical coding.
- Improved **Speech-to-Text** precision with a custom **spell checker**.
- Developed a **patient attendance prediction** model for clinic show-up insights.
- Streamlined information with a **coreference resolution** model.
- Predicted outpatient visit volumes using a **time-series forecasting** model with **Facebook Prophet**.
- Anticipated consultation durations with a **service time prediction** model.
- Orchestrated **deployment** to **AWS ECR** through Dockerization for scalable access.

ACHIEVEMENTS

- **100 percent success rate on upwork with 5 star ratings** from all clients